

June, 1995
Russ Hopping

Pearl Property on Hoveys Pond, West Boxford

Property Description:

The property consists of 2 parcels of contiguous land totaling approximately 8 acres. Both parcels have 250' of frontage on Lake Shore Road to the north. The property abuts Ingalsby West 40, 40 acres farmed by the Price family, to the east and southeast. To the south the property has 25' of frontage along Hoveys Pond. To the west lies developed house lots. The property is flat and is currently under chapter 61 and the majority of it is being cultivated by the Price family. A hedgerow of trees and shrubs outlines the property on all sides except along the western boundary.

Reasons for Protection:

The property contributes to the scenic, ecological, and agricultural values associated with the Greater Hoveys Pond Area. The property contributes to the viewshed as seen from Main Street. Recently the Town of Boxford purchased the Chadwick field and marsh to help preserve this viewshed. The Town of Boxford has also expressed interest in maintaining the last vestiges of agriculture within the Town. This property has traditionally been farmed and is currently being farmed. In terms of its ecological importance the property is part of a contiguous undeveloped agricultural landscape which provides unique and threatened habitat for many species of wildlife. In addition, the property also helps protect the Hoveys Pond watershed. During a recent site walk of the property deer tracks were numerous and an osprey was observed soaring over the property, both testament to how rich and productive this area is for wildlife.

Protection of these 8 acres will help protect and contribute to the Greater Hoveys Pond Conservation Area. Over the years there has been tremendous effort to preserve and maintain the character and integrity of this area. The Town of Boxford in conjunction with the State has invested in protecting this area through the APR program and Self Help Program. More recently private interests have joined in this effort. BTA/BOLT in 1994 purchased 70 acres in the area and remains committed to a Greater Hoveys Pond Conservation Area. The lands associated with the Hoveys Pond area perhaps represent the last opportunity for Boxford to protect and preserve a working, productive landscape which when protected, would provide scenic and recreational opportunities unparalleled to its residents.

Options for protection:

As stated previously, the Town of Boxford has clearly expressed its interest in preserving this area of Town. This property would complement the recent purchase of the Chadwick lands. For this reason this parcel would make an excellent candidate for the self help program. Because of the June deadline for the program, efforts

of protection would need to focus on next years fiscal year. If this method is perused the process should begin as soon as possible since appraisals need to be conducted and paperwork filed. If the property has not been perked, soil tests may need to be done. Although the lots have the necessary road frontage and acreage, it needs to be determined if the lots are buildable under the current regulations. Additional Options may include: payment for property in installments, the purchase of 1 lot at a time, or outright purchase by the Town.

If the Town does purchase this land, the option will remain to lease this land for agricultural purposes. Leasing will offset any loss of taxes that this land contributed since the land has been in chapter 61A.

Russell Hopping 6/95

The property contributes to the ecological and agricultural values associated with the Greater Hoveys Pond Area. The property contributes to the viewshed as seen from Main Street. Recently the Town of Bedford purchased the Chadwick field and marsh to help preserve this viewshed. The Town of Bedford has also expressed interest in maintaining the last vestiges of agriculture within the Town. This property has traditionally been farmed and is currently being farmed. In terms of its ecological importance the property is part of a contiguous undeveloped agricultural landscape which provides unique and threatened habitat for many species of wildlife. In addition, the property also helps protect the Hoveys Pond watershed. During a recent site walk of the property deer tracks were numerous and an osprey was observed soaring over the property. Both testament to how rich and productive this area is for wildlife.

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Options for protection:

As stated previously, the Town of Bedford has clearly expressed its interest in preserving this area of Town. This property would complement the recent purchase of the Quabwick lands. For this reason this parcel would make an excellent candidate for the sale help program. Because of the time deadline for the program, efforts

NR-100000-1000

Thomas G. Farnsworth

BTA/BOLT

PLANT COMMUNITY INVENTORY

of the

HOVEYS POND CONSERVATION AREA

1994

INTRODUCTION

Overview & Purpose:

In the spring of 1994, the Boxford Open Land Trust acquired a 70 acre property in West Boxford, Massachusetts. The Trust purchased the property as a nucleus for a proposed conservation area around Hoveys Pond. In addition, 5 certified vernal pools existed on the property and 3 state listed rare species had been documented from the property as well. However, the property in general was not well known to the Trust. In particular, property boundaries were uncertain. Discussions on managing the property have occurred but lack of baseline information, such as boundaries and plant communities that occur on the property, prevent any reasonable attempt at management.

To the best of the Trust's knowledge, no forestry or wildlife management has occurred on the property in recent times. This property was not listed under the state's forestry tax break program (chapter 61) prior to acquisition. The Conservation Commission files that pertained to this property were reviewed. A wildlife assessment (Appendix A) was conducted for the 50 foot road dedication only for the proposed subdivision at the time the property was before the Conservation Commission. Although this document provides some information on soil types, hydrology and forest history, it is very limited in its scope and therefore the value of its results are suspect and of very little use. Other literature reviewed included: Soil maps from the USDA, The Estimated Habitat Map produced by the Massachusetts Natural Heritage and Endangered Species Program which identifies all suspected rare wetland species habitat. Tree and shrub identification guides were consulted for plant identification.

Property boundaries were located and an inventory of the major (> 1 acre) plant communities was conducted for the purpose of developing an accurate base map with an overlay depicting all the plant communities. Landscape features such as topography and hydrography were also identified for inclusion in the base map. It was suspected that multiple plant communities existed within the 70 acres and that these plant communities

would vary in area covered. These maps and overlays could then be used to aid the Trust in property management (i.e. boundary posting, placement of trails, as well as for any forestry and wildlife management).

The existence of at least 3 rare wetlands species signals caution in moving quickly with property management. Those activities which are identified as non-compatible with these rare species need to be identified. The movement and requirements of these species is poorly known, thus a complete assessment of what exists on the property, as well as a complete understanding of the rare species is essential before any management decisions are implemented. The development of an accurate base map and a plant community inventory are the first necessary steps in the development of a viable management plan for the 70 acres

Description of Study Area and Population:

The study area consists of 70 acres of mostly wooded uplands and forested wetlands. The property lies between Hoveys Pond and Sperrys Pond but does not actually have any frontage on either. The property has poor access and the only frontage on any road is 181.04' along Route 133. Topography is gently slopping to abrupt with rock outcroppings and surface boulders. Soils are mostly Rock outcrop-Charlton-Hollis complexes and deep Medisaprists. The upland forests are primarily mixed deciduous with stands of hemlock. Red maple is the principle tree species of the forested wetlands. Vernal pools (temporary ponds which often dry up in the summer) occur in many of the low spots and between rock outcroppings. Five of the vernal pools have been certified through the Vernal Pool Certification Program administered by the Natural Heritage and Endangered Species Program of Massachusetts. At least 2 of these pools are known to harbor Blanding's Turtle, Spotted Turtle and Blue-spotted Salamanders, all state listed rare species. The western edge of the 70 acres is mostly wetland. An intermittent stream flows from this wetland to Sperrys Pond. A major powerline corridor bisects the property in two

places. The habitat under these powerlines is mostly early successional with herbaceous growth and shrubs.

This property came before the Boxford Conservation Commission in 1992 and 1993 as a site for a cluster housing development. Prior to this in 1990 and 1991 parts of the property along route 133 were subdivided and sold as house lots. Boundary delineation is obscured in part by this fact. This subdivision involved the creation of 3 house lots along the western edge of the 70 acres. The lot boundaries do not follow the existing stone walls and are irregular in configuration. Much of this acreage is shrub swamp or marsh which complicates physical conformation of boundaries.

Historically the land is too rocky, steep or wet to have supported cultivated agriculture. Attempts to drain the wetland in the past are visible. The outflow from the large wetland along the western side of the property has been excavated. Stonewalls are common, juniper and red cedar persist throughout parts of the property, and several large wolf trees occur along the stonewalls of the property boundaries all suggesting that this property was used as pasture. The juniper and cedar are sparse and for the most part, in poor health, persisting mainly in the rockiest and driest areas. The size of the surrounding trees in conjunction with the soil type, suggests the pasture was abandoned probably more than one hundred years ago. The oldest forests on the property show no multiple trunking suggesting no cutting has taken place and that these forests have been relatively undisturbed since the pasture was abandoned. Other parts of the forest suggest there has been at least two separate cuttings with the most recent occurring approximately 20 to 30 years ago.

METHODS & MATERIALS

The Boxford Open Land Trust provided the deed, survey maps and aerial infrared photographs with wetland overlays. The Boxford Conservation Commission provided

aerial black and white photographs. Jim MacDougall, Land Manager for Essex County Greenbelt Association provided technical support for evaluating these resources. Equipment included a Suunto A1000 compass, 50 meter measuring tape, light table, sheets of mylar and transparencies. Additional materials included a USGS topographic map of the area and a copy machine. The property boundaries and plant communities were delineated throughout the months of September and October. The species area curves and plot sampling were carried out throughout the month of November.

Base Map:

Boundaries were located and field checked using the maps and aerial photographs. Scales were converted to 1" = 400' for reproduction of the base map on 9.5 x 11 sheets of paper.

Plant Community Overlay:

Plant communities were determined using aerial color infrared and black and white photographs. Areas that appeared distinctly different in color and texture on these photographs were considered distinct plant communities and outlined on transparency. These areas were then ground truthed for confirmation.

Due to the size of the property and the time constraints associated with an inventory of this size, it is likely that small and inconspicuous plant communities were overlooked. It is also inevitable in an inventory of this type, that rare and uncommon species were not represented in the sampling.

Sampling:

Once plant communities had been determined and delineated they were sampled using a Multiple Plot Method. Due to time constraints only the woody vegetation of the upland communities was sampled. All communities were sampled with 3 plots except the

white pine community in the southeast corner of the property which was sampled using 1 plot because of the large plot size. Plot location was determined randomly using a grid and random number table. Graph paper with 4 squares per inch was used for the grid. Edges of communities were avoided to reduce the edge effect which could influence (bias) the data. Plot size was determined by conducting a species area curve for each community. Within each plot the woody vegetation was sampled according to the guidelines as follows:

Trees = $> 5''$ dbh and $> 20'$ high

Saplings = $> .4''$ to $5''$ dbh and $> 20'$ high

Shrubs = $3'$ to $20'$ high

Seedlings = $< 3'$ high

Within each plot the diameter at breast height (dbh) was recorded and converted to basal area for each tree by species. Saplings were counted and recorded by species. Shrubs were counted and aerial coverage recorded by species. Seedlings were counted by species. Structure and composition of the plant communities were evaluated using the following:

Mean dbh = total dbh per species / total # of trees

Total basal area = sum of basal area per species in square inches

Dominance = total basal area per species / area (ha)

Relative Dominance = total basal area per species / total basal area of all species

Frequency = # of plots where species occurred

Relative Frequency = # of occurrences for species / total # of occurrences of all species

Density = # of trees per species / area (acre)

Relative Density = # of trees per species / total # of trees for all species

Aerial Coverage = radius of shrub squared x pi

Reproduction = the # of saplings and shrubs for each species.

Importance Value = Rel. Dominance + Rel. Frequency + Rel. Density

Upon analysis of the sampling data, the following classifications, preceded by the dominant species for the upland communities, were then applied to the plant communities:

1. Hardwoods = < 10% softwoods
2. Softwoods = < 10% hardwoods
3. Mixed Woods = > 40% of each type
4. Mixed Hardwoods Dominant = 60% - 90% hdwds in canopy
5. Mixed Softwoods Dominant = 60% - 90% sftwds in canopy
6. Wooded Swamp = forested wetlands
7. Shrub Swamp = wetlands dominated by shrubs with sparse to no tree canopy
8. Marsh = wetland dominated by emergent vegetation
9. Early Successional - areas with no tree or shrub layer (i.e. powerlines)

RESULTS

A base map was generated at a scale of 1" = 400' for practical use by the Trust. The plant community overlay was generated at the same scale as the base map and identified 10 plant communities on the 70 acres. Of the 10 communities 6 were sampled. Of the remaining 4 communities 3 were wetlands and 1 was the plant community that occurs under the main powerline corridor at the northern end of the property. This area is early successional and is maintained by the New England Power Company. Tree or sapling vegetation is not characteristic of this community thus it was not sampled. The powerlines along the western edge of the property bisect wetland and are characterized as shrub swamp. The wetland communities had been delineated and characterized previously on

overlays with the color infrared photographs. These delineations were field checked for accuracy and adjustments were made if necessary.

White Pine Forest: This is a mixed softwoods dominant forest located in the southeast corner of the property. White pine (74.75%) is dominant. Red oak (17.43%) and white oak (5.80%) are sub-dominant. Saplings include white pine (2) and red oak (3). Shrubs are dominated by hop hornbeam (63.36%) and white pine (30.08%).

Red Oak / White Pine Forest: This is a mixed hardwoods dominant forest located in the south central part of the property. Red oak (63.15%) is dominant. White pine is sub-dominant (27.69%). Saplings include hop hornbeam (3), yellow birch (1), white ash (1), and red maple (1). The shrub layer is dominated by hop hornbeam (57.59%).

Older Red Oak Forest: This is a hardwoods forest located in the southern and central parts of the property. Red oak (81.73%) is the dominant species. White oak (12.85%) and pignut hickory (5.41%) comprise the remaining species. The only saplings were 2 hop hornbeam. Shrubs are dominated by hop hornbeam (73.05%). White pine (6.98%) and black cherry (5.61%) are sub-dominant.

Red Maple Forest: This is a hardwoods forest located along the southwestern edge of the property. This plant community abuts the powerlines which bisect the western edge of the property. Red maple (98.04%) is dominant. Saplings include white ash (7), pignut hickory (1) and bitternut hickory (1). Spicebush (97.26%) is the dominant shrub.

Hemlock Forest: This is a mixed softwoods dominant forest. Most of this community type occurs along the eastern edge of the property with small stands occurring throughout the northern and central portions of the property as well. Hemlock is the dominant species (84.69%). Saplings include hemlock (2) and yellow birch (1). No shrubs were encountered within the plots.

Secondary Red Oak Forest: This is a hardwoods forest located in the north central portion of the property. Red oak (90.27%) is dominant with white oak (9.73%)

sub-dominant. Saplings included pignut hickory (3), hop hornbeam (2) and red oak (1). Hop hornbeam (45.02%) dominates the shrub layer. Hemlock (22.85%) and maple-leaved viburnum (10.00%) are sub-dominant.

DISCUSSION

As suspected, multiple plant communities were encountered on the 70 acres. As previously mentioned the wetlands plant communities were not sampled because of time constraints and previous delineation and characterization by the State of Massachusetts and the University of Massachusetts. In the future if the Trust decides to manage the wetland areas then a complete inventory with sampling should be conducted.

The plant communities that exist under the powerlines were not sampled as well. No trees or saplings exist under the powerlines, shrubs and herbaceous species are dominant. This area is maintained by the New England Power Company as a powerline corridor and will continue to be maintained for this use in the future. Although this area has wildlife and recreational value, it was not sampled because of the lack of structural diversity of the woody vegetation and management control by the Trust.

This inventory was conducted to identify plant communities and their boundaries. It was also conducted to evaluate the future composition of the forest communities that currently exist on the property. By evaluating the reproduction of species within the plots, it is possible to assess the future composition, provided disturbance patterns do not change.

White Pine Forest: This forest community is dominated by white pine. Red oak is sub-dominant. Both these species are represented by saplings thus, they will continue to be represented in the canopy and likely remain the dominant species for some time. White pine is also well represented in the shrub and seedling layers. Many, if not most of these white pines will not survive due to the shading of the canopy. Red oak is only represented by 1 individual in the shrub layer and seedlings were uncommon. Unless these seedlings

receive increased amounts of sunlight, they also will not survive. Hop hornbeam is the dominant shrub. This is a shade tolerant tree that will likely continue to dominate the shrub layer and the lower understory.

Red Oak / White Pine Forest: Red oak is dominant with white pine sub-dominant. Other species of hardwoods occur including but not limited to: white oak, yellow birch, white ash, red maple and hickories. Neither of the dominant species are represented by saplings. There were a few white pine shrubs and seedling but no red oak shrubs or seedlings. Species that are represented by saplings include: yellow birch, white ash, hop hornbeam and red maple. The shrubs are again dominated by hop hornbeam. It would seem then, without some disturbance which would open up the canopy allowing more light to reach the forest floor, the hardwood species represented by saplings and shrubs will dominate this forest community in the future. Hop hornbeam will likely continue to dominate the shrub and lower understory layers.

Older Red Oak Forest: Red oak is dominant with white oak sub-dominant. Other species include: white oak, shagbark hickory, pignut hickory, white ash and red maple. Neither of the dominant species are reproducing well. There are no saplings for either oak and both species are poorly represented by shrubs. Seedlings for both are uncommon and likely will fail to survive due to the lack of light. The only saplings were hop hornbeam. The shrub layer was dominated by hop hornbeam as well. This forest will likely include more hickories and other hardwoods in time. Hop hornbeam will continue to dominate the understory and shrub layers. This community is referred to as an oak-hickory forest by Dr. Olson in the wildlife report (appendix. A).

Red Maple Forest: This forest is dominated by red maple with white ash sub-dominant. Other species observed outside plots include American elm and along the upper edge, white oak. Red maple is not reproducing at all. No saplings or shrubs were encountered. White ash, however, is the dominant sapling and has a few shrubs as well. Saplings also include both pignut and bitternut hickories. Both hickories also are

represented by a few shrubs. The shrub layer is almost exclusively spicebush. This forest is probably a wetland according to the Massachusetts Wetland Protection Act, although parts of the upper slope edge, are probably not wetland. Since there is no red maple reproduction, this forest will likely become dominated by white ash and hickories in the future. Spicebush will likely remain the dominant shrub provided the water level of the nearby shrub swamp and marsh remains constant.

Hemlock Forest: This forest is dominated by hemlock. Some white oak, red oak, white ash, red maple and hickories occur throughout. Although red maple has a dominance of 11%, all of the individuals were recorded from one plot which happened to be located in a damp area. I do not believe this is a true representation of the dominance of this species throughout the hemlock forest. Hemlock and yellow birch were the only saplings recorded. No shrubs were recorded and very few seedlings. The lack of understory vegetation can be explained by the lack of light penetrating the dense hemlock canopy. Because of this shading and because hemlock saplings exist, this forest community will doubtless persist well into the future. Hemlock is also a long lived species, capable of surviving for many hundreds of years. This community is also expanding. Many shrub and sapling sized hemlocks exist along the edge of the hemlock community and the neighboring hardwoods communities.

Secondary Red Oak Forest: This forest is dominated by red oak. White oak is a distant sub-dominant. Other species include white birch and pignut hickory. Saplings were recorded for red oak, pignut hickory and hop hornbeam. Pignut hickory dominated the saplings. Both species of oak were represented in the shrub layer but together their relative dominance equaled less than 7%. Hop hornbeam and hemlock dominate the shrub layer. Although the oaks are reproducing, white oak is reproducing poorly. However, this forest community is young and many of the trees are multiple trunked, evidence of cutting approximately 20 to 30 years ago. The oaks will maintain their dominance well into the future. Pignut hickory will likely become established in the canopy as well. However, the

second most dominant species in the shrub layer is hemlock. Hemlock forest borders the oak forest community to the east and small stands occur around it elsewhere. Hemlock is likely to continue to colonize and eventually dominate this forest community.

Additional Species Information:

Although only 1 beech was recorded during the sampling, it does exist throughout the property. This species is not common but does occur in several very small stands. However, many sapling and shrub sized beech were observed within the oak forests and along the edges of the hemlock forest. This is a shade tolerant species that is likely to become more prevalent over time.

An uncommon but very site specific species, black gum was observed around some of the wetlands edge. White swamp oak was very site specific around several of the vernal pools. Musclewood occurred in one location adjacent to a wetland on the southwestern side of the property. A small stand of mountain laurel occurs here as well. Witchhazel is found around the slopes of a small wetland pool along the southwestern boundary only. Other rarities included American linden (basswood) and American chestnut.

SUMMARY & CONCLUSION

Ten communities were identified on the 70 acres in West Boxford. Six upland forested communities were sampled. Red oak is the dominant hardwood on the property but reproduction is poor. Red maple, white ash, hickories and other hardwoods are likely to increase their dominance in the future. Hemlock is the dominant softwood with good reproduction. White pine is dominant only over a small area. Softwoods, both hemlock and white pine, are colonizing the hardwoods forests in places. This trend is likely to continue barring future disturbance. Hop hornbeam is by far the most dominant understory

species. Since hop hornbeam is a shade tolerant species it is likely to persist as the dominant species of the understory and shrub layers throughout.

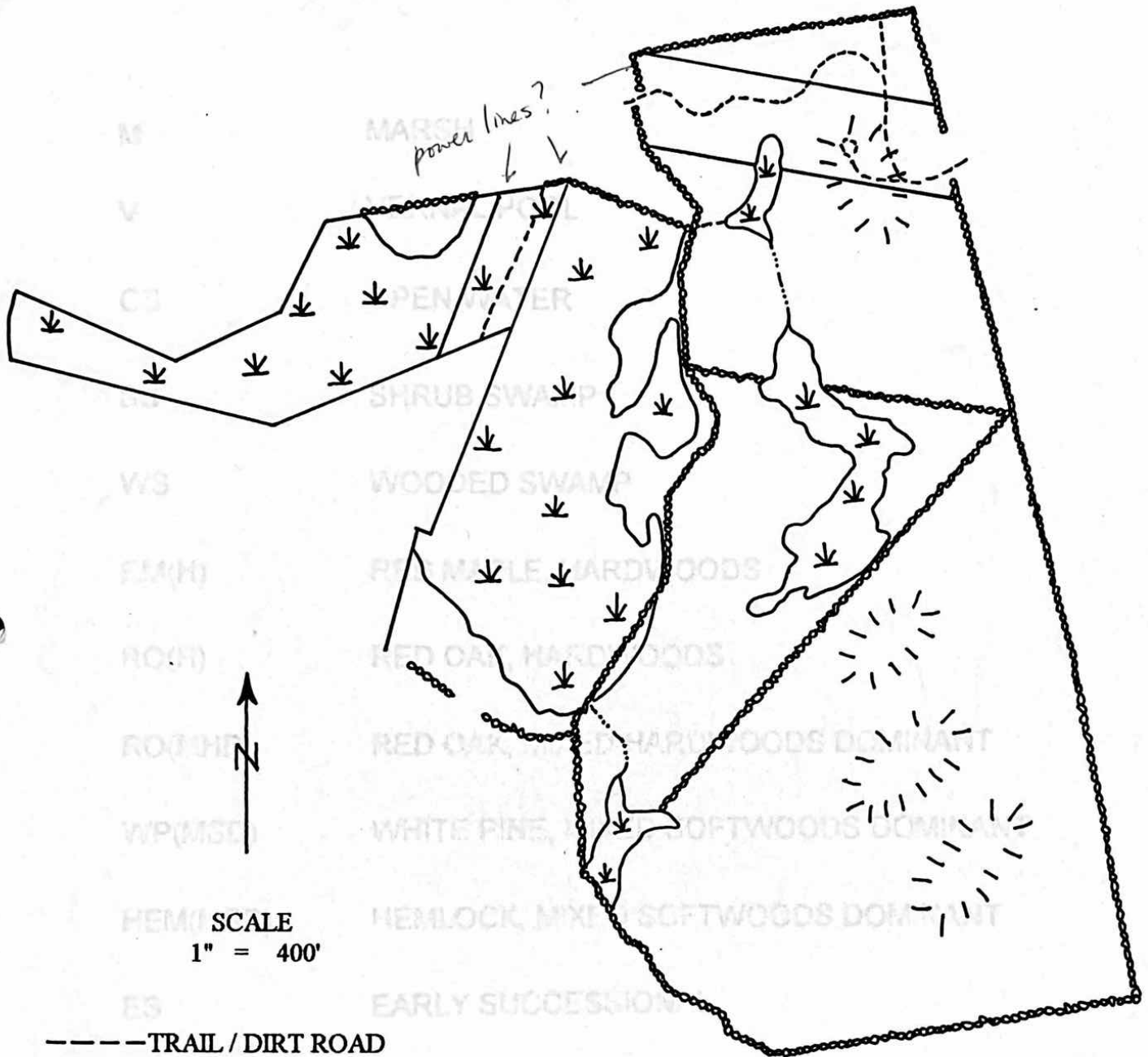
Evidence suggests the plant communities have changed dramatically over the last 100 years or so. This area was originally cleared and utilized as pasture. The pasture was eventually abandoned and allowed to revert back to forest. Signs of pasture still remain as evidenced by the red cedar and juniper. The current forest communities have developed over the last 50 to 100 years. It is likely these communities will continue to change without forest management or other disturbances.

This inventory attempted to identify the major plant communities that exist on the BTA/ BOLT property and how these communities will likely change in the future. It is my feeling that this inventory accomplished what it set out to do and the results will prove valuable to the Trust. However, although the sampling produced a decent analysis of the existing communities, statistically it is weak. If time and resources had allowed, more plots per community should have been used. Also smaller community types exist within the larger communities. These communities were not sampled because of time and resource constraints. If the Trust feels that it would be valuable to inventory for these communities they could easily be added to the base map and overlay in the future.

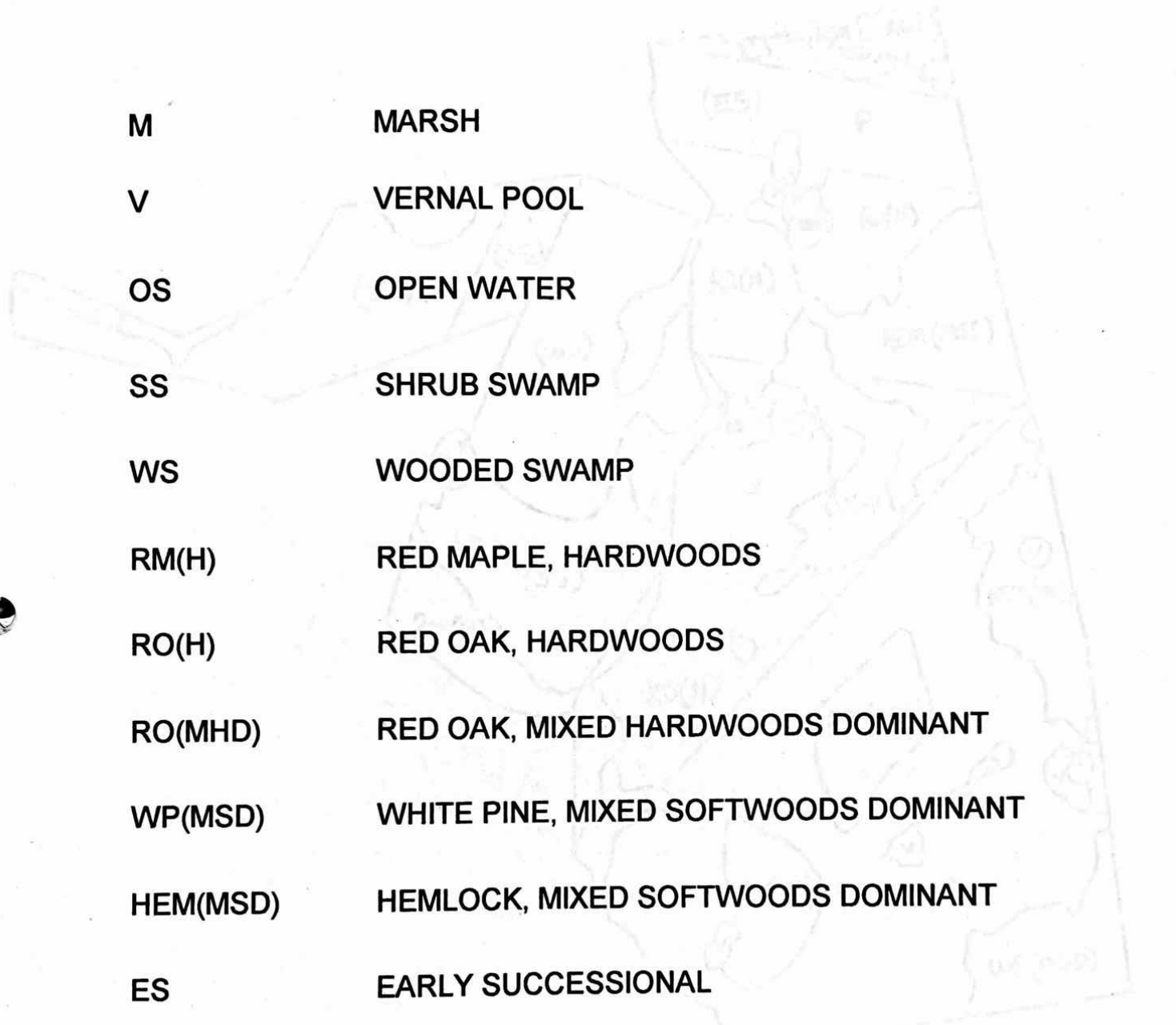
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Boxford Open Land Trust
West Boxford

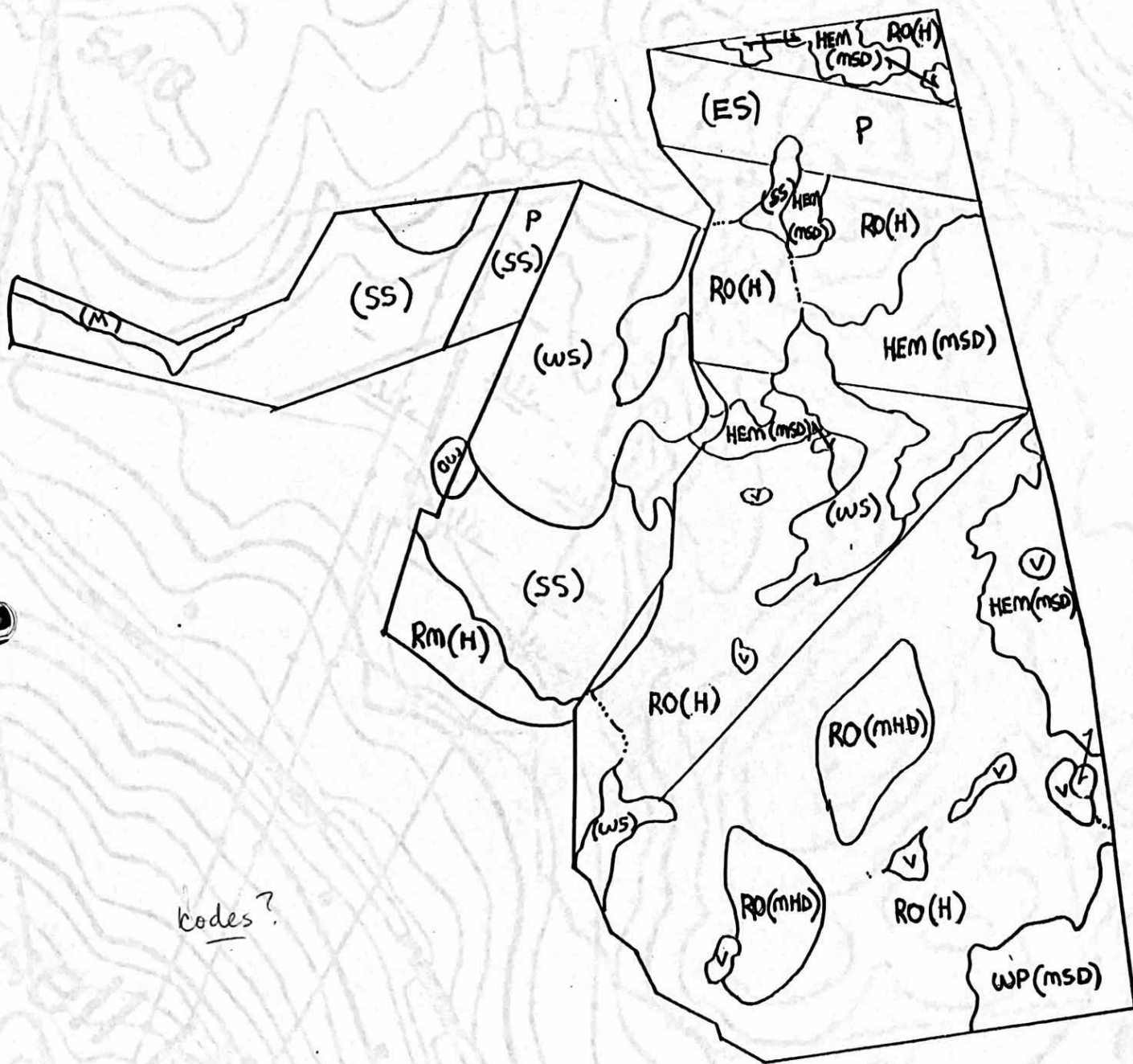


PLANT COMMUNITIES

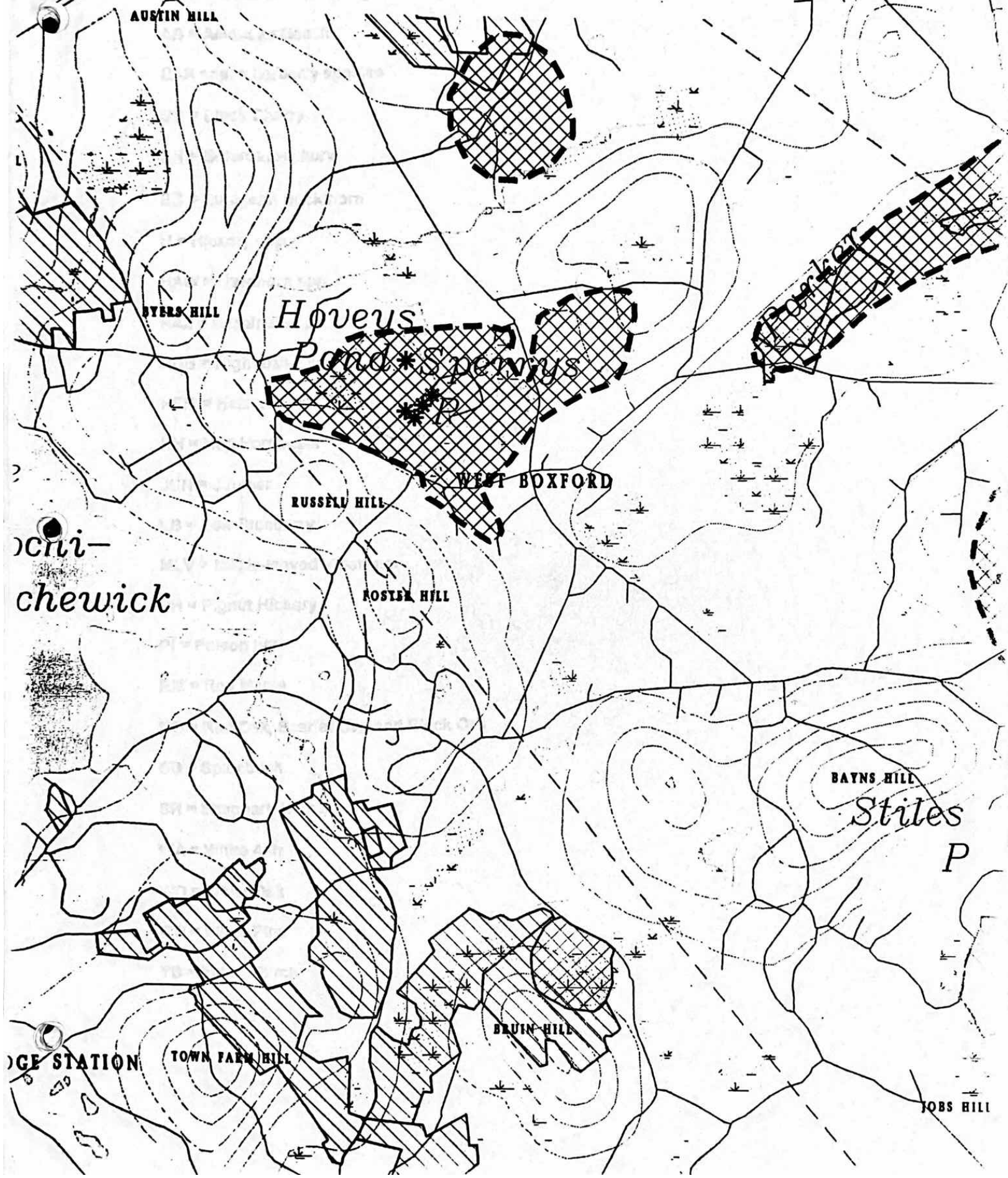


M	MARSH
V	VERNAL POOL
OS	OPEN WATER
SS	SHRUB SWAMP
WS	WOODED SWAMP
RM(H)	RED MAPLE, HARDWOODS
RO(H)	RED OAK, HARDWOODS
RO(MHD)	RED OAK, MIXED HARDWOODS DOMINANT
WP(MSD)	WHITE PINE, MIXED SOFTWOODS DOMINANT
HEM(MSD)	HEMLOCK, MIXED SOFTWOODS DOMINANT
ES	EARLY SUCCESSIONAL
P	POWERLINES

PLANT COMMUNITIES



ESTIMATED HABITAT



Species List

ABAR = American Barberry

AB = American Beech

BAR spp. = barberry species

BC = Black Cherry

BH = Bitternut Hickory

EB = European Buckthorn

H = Hickory spp.

HAW = Hawthorn spp.

HAZ = Hazelnut spp.

HBB = Highbush Blueberry

HEM = Hemlock

HH = Hop Hornbeam

JUN = Juniper

LB = Low Blueberry

MLV = Maple-leaved Viburnum

PH = Pignut Hickory

PI = Poison Ivy

RM = Red Maple

RO = Red Oak, Scarlet Oak and Black Oak

SB = Spicebush

SH = Shagbark Hickory

WA = White Ash

WO = White Oak

WP = White Pine

YB = Yellow Birch

Summary of Plot Sampling for Trees and Saplings

Species	# of Plots	Mean DBH	Total PA	Rel. Frequency	Rel. Density	Rel. Volume
WBP	5	18.5	1500.00	74.75%	81.67%	2
RM	1	7.2	60.72	2.07%	6.33%	2
RO	4	10.1	404.44	37.47%	39.33%	2
WO	2	11.75	118.0	5.37%	10.47%	2

Summary of Plot Sampling for Trees and Saplings

Date: 11 - 25 - 94 Location: BTA / BOLT 70ac. West Boxford

of Plots: 1 Area Sampled: White Pine Forest (10 x 32) = 320m2

Community Description: Mixed Softwoods Dominant. White pine dominant. White pine and hop hornbeam dominant understory.

Species	# of Indiv.	Mean DBH	Total BA	Dominance	Rel. Dominance	Frequency	Rel. Frequency	Density	Rel. Density	Impor. Value	Saplings
WP	5	18.5	1506.66	17830 / ha	74.75%			156.25	41.67%		2
RM	1	7.2	40.72	482 / ha	2.02%			31.25	8.33%		
RO	4	10.1	351.31	4158 / ha	17.43%			125	33.33%		3
WO	2	11.75	116.9	1383 / ha	5.80%			62.5	16.67%		
4	12		2015.59	23853	100			375	100		5

Summery of Plot Sampling for Shrubs

Date: 11 - 25 - 94 Location: BTA / BOLT 70ac. West Boxford

of Plots: 1 Area Sampled: White Pine Forest (10 x 32) = 320m²

Community Description: Mixed Softwoods Dominant. White pine dominant. White pine and hop hornbeam dominant understory.

Species	# of Indiv.	Coverage	Rel. Dominance	Freqeency	Rel. Frequency	Rel. Density	Impor. Value
WP	12	24538	30.08%			25.53%	
RM	10	1980	2.43%			21.28%	
EB	2	904	1.11%			4.26%	
BC	2	565	0.69%			4.26%	
WO	2	904	1.11%			4.26%	
HH	14	51683	63.36%			29.79%	
RO	1	113	0.14%			2.13%	
WA	1	113	0.14%			2.13%	
HBB	1	452	0.55%			2.13%	
MLV	2	314	0.38%			4.26%	
10	47	81566	100%			100%	

PLOT # 1 WHITE PINE FOREST

DATE 11-25-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
WP	21.7	369.8	2	(12)	C	
WP	21.9	376.7				
WP	26.0	530.9				
WP	7.6	45.36				
WP	15.3	183.9				
Rm	7.2	40.72		(10)		40.72
RO	5.3	22.06	3	(1)	U	
RO	11.0	95.03				
RO	10.1	80.12				
RO	14.1	156.1				
WO	11.7	107.5		(2)	R	116.9
WO	11.8	109.4				
BC				(2)		
WA				(1)		
HH				(14)		
HBB				(1)		
MLV				(2)		
EB				(2)		
JUN					F	
LB					A	
PH					R	
ABAR					R	

A = Abundant + 50

C = Common 25 - 49

F = Fairly common 10 - 24

U = Uncommon 3 - 9

R = Rare 1 - 2

Summary of Plot Sampling for Trees and Saplings

Date: 11 - 26 - 94 Location: BTA / BOLT 70 ac. West Boxford

of Plots: 3 Area Sampled: Red Oak / White Pine Forest (5.5 x 11.5) = 63.25m²

Community Description: Mixed Hardwoods Dominant Red oak is dominant canopy species. Hop hornbeam is dominant in understory.

Species	# of Indiv.	Mean DBH"	Total BA	Dominance	Rel. Dominance	Frequency	Rel. Frequency	Density	Rel. Density	Impor. Value	Saplings
RO	5	13.26	570.15	9014 / ha	63.15%	3	33.33%	105.4	45.45%	141.93	
WP	3	13.37	250	3953 / ha	27.69%	3	33.33%	63.24	27.27%	88.29	
YB	1	9.1	65.04	1028 / ha	7.20%	1	11.11%	21.08	9.09%	27.4	1
WO	1	10.1	7.5	119 / ha	0.83%	1	11.11%	21.08	9.09%	21.03	
PH	1	7.5	10.1	160 / ha	1.12%	1	11.11%	21.08	9.09%	21.32	
WA											1
HH											3
RM											1
8	11		902.79	14274	100	9	100	231.88	100	300	6

Summary of Plot Sampling for Shrubs

Date: 11 - 26 - 94 Location: BTA / BOLT 70ac. West Boxford

of Plots: 3 Area Sampled: Red Oak / White Pine Forest (5.5 x 11.5) = 63.25m²

Community Description: Mixed Hardwoods Dominant. Red oak is dominant canopy species. Hop hornbeam is dominant in understory.

Species	# of Indiv.	Coverage	Rel. Dominance	Frequency	Rel. Frequency	Rel. Density	Impor. Value
WP	4	6332	8.86%	2	10%	6.45%	25.31
YB	2	3618	5.06%	1	5%	3.22%	13.28
WO	6	5653	7.91%	3	15%	9.68%	32.59
PH	1	28	0.04%	1	5%	1.61%	6.65
WA	3	4212	5.89%	2	10%	4.84%	20.73
HH	26	41156	57.59%	3	15%	41.94%	114.53
RM	4	4183	5.83%	2	10%	6.45%	22.28
BC	5	2688	3.76%	3	15%	8.06%	26.82
MLV	2	56	0.08%	1	5%	3.22%	8.3
HAZ	3	1017	1.42%	1	5%	4.84%	11.26
HHB	6	2515	3.52%	1	5%	9.68%	18.2
11	62	71458	100%	20	100%	100%	300%

PLOT # 1 RED OAK / WHITE PINE FOREST DATE 11-25-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
RO	20.0	314.2				314.2
WP	5.4	22.90		(1)		22.90
YB	9.1	65.04				65.04
WA			1	(1)		
HH			1	(14)	F	
RM			1	(2)		
BC				(3)	U	
WO				(1)	R	
HAZ				(3)		
LB					F	
LS					F	

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

PLOT# 2 RED OAK/WHITE PINE FOREST DATE 11-26-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
RO	11.8	109.36				
RO	12.6	124.69				254.05
WP	16.3	208.7		(3)	U	208.7
HH			1	(4)	R	
YB			1	(2)		
BC				(1)		
WO				(1)	F	
WA				(2)		
PH				(1)		
JUN					R	
LB					F	

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

PLOT # 3 RED OAK / WHITE PINE FOREST

DATE 11-26-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
RO	12.4					
RO	9.5					21.1
WO	10.1			(4)	U	10.1
PH	7.5				R	7.5
WP	18.4					18.4
HH			1	(8)	R	
MLV				(2)	R	
Rm				(2)		
BC				(8)	R	
HBB				(6)		
JON					R	
LB					R	
					RO	512.15
					WO	250.0
					PH	7.5
					PH	7.5
					PH	7.5
					WO	10.1
						702.71

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

Summary of Plot Sampling for Trees and Saplings

Date: 11 - 26 - 94 Location: BTA / BOLT 70 ac. West Boxford

of Plots: 3 Area Sampled: Older Red Oak Forest (5.5 x 11.5) = 63.25m²

Community Description: Hardwoods. Red oak dominant with white oak and pignut hickory. Hop hornbeam dominant in understory

Species	# of Indiv.	Mean DBH"	Total BA	Dominance	Rel. Dominance	Frequency	Rel. Frequency	Density	Rel. Density	Impor. Value	Saplings
WO	2	9.25	134.72	2130 / ha	12.85%	2	28.57%	42.16	15.38%	56.8	
RO	9	10.69	856.65	13544 / ha	81.73%	3	42.86%	189.72	69.23%	193.82	
PH	2	8.8	56.75	897 / ha	5.41%	2	28.57%	42.16	15.38%	49.36	
HH											2
WP											
MLV											
SC											
WA											
SH											
HAZ											
4	13		1048.12	16571	100	7	100	274.04	100	300	2

Summary of Plot Sampling for Shrubs

Date	11 - 26 - 94	Location	BTA / BOLT 70ac. West Boxford
# of Plots	3	Area Sampled	Older Red Oak Forest (5.5 x 11.5) = 63.25m ²
Community Description	Hardwoods.	Red oak dominant with white oak and pignut hickory.	Hop hornbeam dominant in understory.

Species	# of Indiv.	Coverage	Rel. Dominance	Frequency	Rel Frequency	Rel. Density	Impor. Value
WO	2	226	0.25%	1	5.88%	2.99%	9.12
RO	2	1922	2.19%	1	5.88%	2.99%	11.06
HH	17	66267	73.05%	2	11.76%	25.37%	110.18
RM	11	2294	2.53%	2	11.76%	16.42%	30.71
WP	3	6332	6.98%	2	11.76%	4.48%	23.22
MLV	21	1325	1.46%	3	17.65%	31.34%	50.45
BC	4	5087	5.61%	2	11.76%	5.97%	23.34
WA	4	2725	3.00%	2	11.76%	5.97%	20.73
SH	2	4083	4.50%	1	5.88%	2.99%	13.37
HAZ	1	452	0.50%	1	5.88%	1.49%	7.87
10	67	90713	100%	17	100%	100%	300

PLOT # 1 OLDER RED OAK FOREST

DATE 11-26-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
WO	9.7	73.90		(2)	U	
WO	8.8	60.82				134.72
RO	15.7	193.6			U	
RO	10	78.54				772.14
RM				(9)	U	
WP				(2)	R	
MLV				(1)		
JUN					R	
PH					R	
HH					U	
LB					A	

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

PLOT# 2 OLDER RED OAK FOREST DATE 11-26-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
RO	11.5	103.87				
RO	11.3	100.29				
RO	10.0	78.54				252.7
PH	8.5	56.75				56.75
HH		25.04	1	(8)	U	
BC				(4)	U	
WP				(1)		
WA				(3)		
SH				(2)		
MLV				(6)	U	
RM					R	
JUN					R	
WO					U	
PH					R	
ABAR					R	

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

PLOT# 3 OLDER RED OAK FOREST DATE 11-26-94

SURROUNDING AREA DESCRIPTION

[illegible]

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

Summery of Plot Sampling for Trees and Saplings

Date: 11 - 26 - 94 Location: BTA / BOLT 70 ac. West Boxford

of Plots: 3 Area Sampled: Red Maple Forest (5.5 x 11.5) = 63.25m²

Community Description: Hardwoods. Red maple dominant with a few white ash Spicebush and hickory understory.

Species	# of Indiv.	Mean DBH"	Total BA	Dominance	Rel. Dominance	Frequency	Rel. Frequency	Density	Rel. Density	Impor. Value	Saplings
RM	9	15.4	1764.07	27890 / ha	98.04%	3	60.00%	189.72	90.00%	248.05	
WA	1	6.7	35.26	557 / ha	1.96%	2	40.00%	15.78	10.00%	51.96	7
PH											1
BH											1
4	10		1799.33	28447	100	5	100	205.5	100	300	9

Summery of Plot Sampling for Shrubs

Date: 11 - 26 - 94 Location: BTA / BOLT 70ac. West Boxford

of Plots: 3 Area Sampled: Red Maple Forest (5.5 x 11.5) = 63.25m2

Community Description: Hardwoods Red Maple dominant with a few white ash. Spicebush and hickory understory.

Species	# of Indiv.	Coverage	Rel. Dominance	Fregencey	Rel Frequency	Rel. Density	Impor. Value
WA	3	467	0.56%	2	16.67%	6.82%	24.05
SB	32	81558	97.26%	3	25%	72.73%	194.99
WO	2	565	0.67%	2	16.67%	4.55%	21.89
ABAR	1	452	0.54%	1	8.33%	2.27%	11.14
HBB	2	565	0.67%	1	8.33%	4.55%	13.35
PH	1	113	0.13%	1	8.33%	2.27%	10.73
BH	2	24	0.03%	1	8.33%	4.55%	12.91
BC	1	113	0.13%	1	8.33%	2.27%	10.73
Σ	10	83857	100%	12	100	100%	300%

PLOT # | RED MAPLE FOREST

DATE 11-26-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
Rm	18.9	280.6				
Rm	12.9	130.70				411.5
WA	6.7	35.26	3	(1)		22.00
SB			1	(16)		
WO				(1)		1
ABAR				(1)	U	
H spp.					R	
PI			1		F	
HBB			1		R	
RO					R	

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

DATE 11-26-94

[illegible]

U = Uncommon 3 - 9 R = Rare 1 - 2

DATE 11-26-94

SURROUNDING AREA DESCRIPTION

[illegible]

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

Summery of Plot Sampling for Trees and Saplings

Date: 11 - 26 - 94 Location: BTA / BOLT 70 ac. West Boxford
 # of Plots: 3 Area Sampled: Hemlock Forest $(6.5 \times 13) = 84.5 \text{ m}^2$
 Community Description: Mixed Softwoods Dominant. Hemlock dominant canopy species. Hickories, oaks and red maple occur as well.
 Yellow birch, hickories and hop hornbean make up sparse understory.

Species	# of Indiv.	Mean DBH"	Total BA	Dominance	Rel. Dominance	Frequency	Rel. Frequency	Density	Rel. Density	Impor. Value	Saplings
HEM	12	15.08	2218.71	26257 / ha	84.69%	3	42.86%	189.3	63.16%	190.71	2
RO	3	7.03	117.81	1394 / ha	4.50%	3	42.86%	43.3	15.79%	63.15	
RM	4	9.25	283.24	3352 / ha	10.81%	1	14.29%	63.1	21.05%	46.15	
YB											1
4	19		2619.76	31003	100	7	100	295.7	100	300	3

PLOT# 1 HEMLOCK FOREST

DATE 11-25-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
HEM	12.8	128.7	1			
HEM	13.0	132.7				
HEM	17.8	248.8				510.2
RO	7.5	44.18				44.18
LB	15.2	31.1			U	
	6.0	18.27				
			1			

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

DATE 11-25-94

SURROUNDING AREA DESCRIPTION

PLOT # 3 HEMLOCK FOREST

DATE 11-26-94

SURROUNDING AREA DESCRIPTION

[illegible]

A = Abundant + 50

C = Common 25 - 49

F = Fairly common 10 - 24

U = Uncommon 3 - 9

R = Rare 1 - 2

Summary of Plot Sampling for Trees and Saplings

Date: 11 - 26 - 94 Location: BTA / BOLT 70 ac. West Boxford

of Plots: 3 Area Sampled: Secondary Red Oak Forest $(5.5 \times 11.5) = 63.25 \text{ m}^2$

Community Description: Hardwoods. Red oak dominant with white oak and hickories. Understory is dominated by hop hornbeam

Species	# of Indiv.	Mean DBH"	Total BA	Dominance	Rel. Dominance	Frequency	Rel. Frequency	Density	Rel. Density	Impor. Value	Saplings
RO	16	9.34	1129.73	17861 / ha	90.27%	3	60.00%	337.28	84.21%	234.48	1
WO	3	7.17	121.79	1926 / ha	9.73%	2	40.00%	63.24	15.79%	65.52	
PH											3
HH											2
4	19		1251.52	19787	100	5	100	400.52	100	300	6

Summery of Plot Sampling for Shrubs

Date: 11 - 26 - 94 Location: BTA / BOLT 70ac. West Boxford

of Plots: 3 Area Sampled: Secondary Red Oak Forest (5.5 x 11.5) = 63.25m²

Community Description: Hardwoods. Red oak dominant with white oak and hickories. Understory is dominated by hop hornbeam.

Species	# of Indiv.	Coverage	Rel. Dominance	Freguency	Rel Frequency	Rel. Density	Impor. Value
PH	4	1130	3.36%	2	11.76%	5.19%	20.31
HH	8	15150	45.02%	3	18%	10.39%	73.06
MLV	43	3350	10.00%	3	17.65%	55.84%	83.49
HAW	6	1698	5.05%	2	11.76%	7.80%	24.61
SH	1	1809	5.38%	1	5.88%	1.30%	12.56
WP	1	452	1.34%	1	5.88%	1.30%	8.52
HEM	5	7688	22.85%	1	5.88%	6.50%	35.23
RM	1	113	0.34%	1	5.88%	1.30%	7.52
AB	1	113	0.34%	1	5.88%	1.30%	7.52
WO	2	565	1.68%	1	5.88%	2.60%	10.16
RO	5	1582	4.70%	1	5.88%	6.50%	17.08
11	77	33650	100%	17	100%	100%	300%

PLOT # 1 SECONDARY RED OAK FOREST DATE 11-25-94

SURROUNDING AREA DESCRIPTION RED OAK DOMINATED WITH WHITE BIRCH, HICKORY,
WHITE OAK SUB DOMINANT WITH HOP HORNBEEAM UNDERSTORY

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
RO	10.8	91.61				
RO	10.6	88.25				
RO	10.8	91.61				
RO	6.1	29.22				
RO	10.9	93.31				304.0
WO	7.9	49.02				49.02
PH	10.2	81.71	1			
HH	9.4	69.40	2	(3)		
MLV				(14)	F	
HAW				(4)	U	
LB					C	
MLV					F	
HAW						
RM						
AB						
PH						
LB					A	
WO					R	

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

PLOT # 2 SECONDARY RED OAK FOREST DATE 11-25-94

SURROUNDING AREA DESCRIPTION

SPECIES	DBH	BA	SAPLINGS	SHRUBS	SEEDLINGS	
RO	9.3	67.93			U	
RO	6.6	34.21				
RO	12.4	120.76			U	
RO	10	78.54				
RO	8.4	55.42				
RO	8.5	56.75				
RO	10.2	81.71				
RO	9.4	69.40			C	564.14
SH				(1)		
HH				(2)	U	
WP				(1)	R	
MLV				(2)	F	
HEM				(5)		
RM				(1)		
AB				(1)		
PH				(2)		
LB					A	
WO					R	

A = Abundant + 50 C = Common 25 - 49 F = Fairly common 10 - 24

U = Uncommon 3 - 9 R = Rare 1 - 2

APPENDIX A

**WILDLIFE STUDY
of
LOT 4A, WASHINGTON STREET
BOXFORD, MA**

**Prepared for:
Washington Street Realty Trust**

**Prepared by:
Daylor Consulting Group, Inc.
313 Congress Street
Boston, MA 02210**

24 September 1992

INTRODUCTION

As a result of impact to wetlands and wildlife resources by proposed road construction at the Washington Street site the Daylor Consulting Group was contracted to investigate the wildlife resources of the area, determine potential impacts to wildlife, and propose mitigation and restoration measures to minimize any possible adverse impacts. This site is included within an area mapped as Estimated Habitats of State-Listed Rare Wetlands Wildlife, 1992 Edition. This work was performed by Kurt N. Olson, Ph.D.

METHODOLOGY

On Tuesday, 22 September I visited the site to determine the level of usage and functional value of the area for wildlife. Particular attention was given to the wetland crossings of the proposed road.

The weather started out as light rain, becoming quite warm and humid as the day progressed. Observations took place between 10 AM and 3 PM.

The procedure and methodology used to investigate these wildlife values involved generally following the guidelines of the Method for the Comparative Evaluation of Nontidal Wetlands in New Hampshire, a commonly used method of functional analysis of wetlands. Also used was the procedure described in New England Wildlife: Management of Forested Habitats, a method developed by the U.S. Forest Service for inventorying wildlife and prescribing management prescriptions.

Field investigative techniques involved walking through each area to be impacted, observing the juxtapositional relationships and composition of cover types, and observing wildlife usage of the areas. Direct wildlife observations included tracks, other signs, rolling over downed tree sections and boulders, and sitting for ten minute periods at each wetland crossing site to listen for and sight wildlife, especially birds.

SITE

The area to be impacted by the proposed road traveling from Washington Street towards Plummer Road first drops down a moderately steep grade of Charlton very stony fine sandy loam. The slope flattens out well before the first wetland crossing where the soil changes to Ridgebury and Leicester extremely stony fine sandy loam. At the intermittent stream a muck soil approximately 12 inches thick overlies an impervious layer of rock. After crossing this wetland a steep bank rises to an

upland area composed of Charlton-Hollis-Rock outcrop soil complexes. At the second wetland crossing the soil becomes a peat 12 inches deep. Depth of inundation during high water as observed from moss lines and other vegetative indicators is about one foot. Both wetlands were dry at the time of observations and no water was observed in holes augered to 20 inches. Between the second wetland and Plummer Road the land rises up a moderately steep stony slope composed of Charlton-Hollis-Rock outcrop soil complexes.

Cover types following the same route begin with a hardwood forest stand dominated by red maple. At the first stream crossing there is a sparse overstory of red maple and white ash and a scrub-shrub wetland dominated by spicebush and speckled alder. Crossing into the upland the cover type becomes an oak-hickory forest. This cover type dominated to the second wetland crossing where a scrub-shrub wetland occurs. Beneath a sparse overstory of surrounding red oak and red maple the dominant shrub species are sweet pepperbush, highbush blueberry, and elderberry. Small open areas occur in this wetland which are covered by pure stands of water buttercup.

Investigations of the wooded areas found no stumps or other evidence of past cutting. This coupled with the lack of old aged trees and intolerant shrub species in the understory including red cedar, juniper, and buckthorn leads me to the conclusion that this area has only recently, ca. 50 years, reverted to forest from agricultural land or pasture.

The first wetland drains off-site southerly to a wetland complex including Sperrys Pond and provides a travel corridor to a wetland at the northerly portion of the property. Immediately off-site to the south a stonewall, stone culvert and fieldstone causeway act to partially dam flowage through this wetland. This also acts as an impediment to travel for reptiles, amphibians and small mammals. The wetland to the south includes shallow marsh immediately off-property. Both of these wetlands are becoming invaded with purple loosestrife which will degrade their value for many species of wildlife.

The second wetland drains off-site through a driveway culvert in an ephemeral stream to the Sperry Pond wetland.

Peat, red oak, open pasture
OBSERVATIONS

Direct observations, visual or hearing, were made of gray squirrel, chipmunk, redback salamander, leopard frog, bluejay, and Canada geese (off-site to north). Indirect observations, tracks, scat, and other sign were made of deer, raccoon, and

skunk. Late summer at mid-day is not the best time to observe wildlife, but these observations do provide an indication of the general types of wildlife utilizing the area.

In the area affected by the proposed road construction there are no good cavity trees, exposed perches or other specialized habitat components available.

CONCLUSIONS

The greatest importance of the wetlands where the road crossings are proposed is their function as travel corridors between Sperrys Pond and other wetlands in the vicinity.

PROPOSED MITIGATION AND REPLICATION

REPLICATION: Three replication areas are proposed. All areas will be planted with shrub species to replicate shrub-shrub wetlands lost through construction.

The two areas at wetland crossing 1 will be planted with spice bush and speckled alder. The replication area at wetland 2 will be planted with spicebush, highbush blueberry, and elderberry. All plantings will be made on five foot centers in three by three blocks. Plant size shall be three feet high and of local genotypes if available.

All organic soils excavated during the construction activities in the wetlands will be stockpiled and utilized in the replication areas. These soils will be kept moist during the period of stockpiling to prevent degradation of the seed bank.

The replication areas will be excavated one foot overdepth and the stockpiled organic soil spread evenly to a depth of one foot. An undisturbed berm of original soil and plant material will be maintained between the original wetland and the replication area shall be maintained. This berm will be breached at the upstream and downstream ends of the replication areas to insure hydrologic continuity with the original wetlands. The area of each breaching will be four feet in length to prevent plugging by debris or vegetative growth.

The replication areas should be monitored annually and reports submitted for two growing seasons to insure that performance standards are met.

MITIGATION: As stated in the conclusion section the most important function of these wetland areas is to provide a travel corridor for wildlife between the major wetlands in the vicinity. As stated earlier there is special importance to these areas due

to the reported occurrence of rare wetland wildlife species, i.e., blue spotted salamander and Blandings turtle. To insure the continued access between these wetlands along these corridors we recommend that instead of using reinforced concrete pipe for culvert that pipe arches or bottomless box culverts be used. This will provide a natural substrate bottom which will much more readily be accepted by wildlife. Culvert size should be large enough to provide passage for raccoon and fox size animals. Steep headwall of 1:1 or steeper slope should extend on 30 degree angles away from the upstream and downstream ends of the culverts to act as funnels forcing wildlife through the culverts rather than crossing the road.

DeGraaf, R.M., et al. 1992. New England Wildlife Management of Wetland Habitats. Gen. Tech. Rep. NE-144. Radnor, PA: USDA Forest Service, NEFS.

DeGraaf, R.M. & D.D. Rudis. 1986. New England Wildlife Management. Natural history and distribution. Gen. Tech. Rep. NE-144. Radnor, PA: USDA Forest Service, NEFS.

RELATIVE ABUNDANCE OF NEW ENGLAND RARE, THREATENED, AND ENDANGERED SPECIES

HABITAT. Wetland, swamp, or forest area (Mason 1954). Occurrence is common in swamps, bogs, and wetlands. It may be found in forested areas, but is more common in open areas. It is found in the wetlands of the New England region (Anderson and Gorman 1987). It is found in the wetlands of the New England region (Anderson and Gorman 1987). It is found in the wetlands of the New England region (Anderson and Gorman 1987).

SPECIAL HABITAT REQUIREMENTS: Ponds or semi-permanent water bodies.

ACCOMPLISHING THE REQUIREMENTS: Good to excellent of 47 to 50 (Mason 1954).

REPRODUCTION: Breeding occurs in ponds or semi-permanent water bodies. Eggs are laid in the water.

REPRODUCTION: Breeding occurs in ponds or semi-permanent water bodies. Eggs are laid in the water. The eggs are laid in the water and the larvae develop in the water.

TIME TO HATCHING: 1 month (Smith 1961:38).

LARVAL PERIOD: Larvae to late June or mid-August (Smith 1961:38).

HOME RANGE/MOVEMENT: Unreported.

FOOD HABITS/PREY: Amphibians, insects, and other small invertebrates.

COMMENTS: A species of conservation interest. It is found in the wetlands of the New England region.

KEY REFERENCES: Anderson and Gorman 1987; Smith 1961; Mason 1954.

OTHER REFERENCES: Anderson and Gorman 1987; Smith 1961; Mason 1954.

OTHER REFERENCES: Anderson and Gorman 1987; Smith 1961; Mason 1954.

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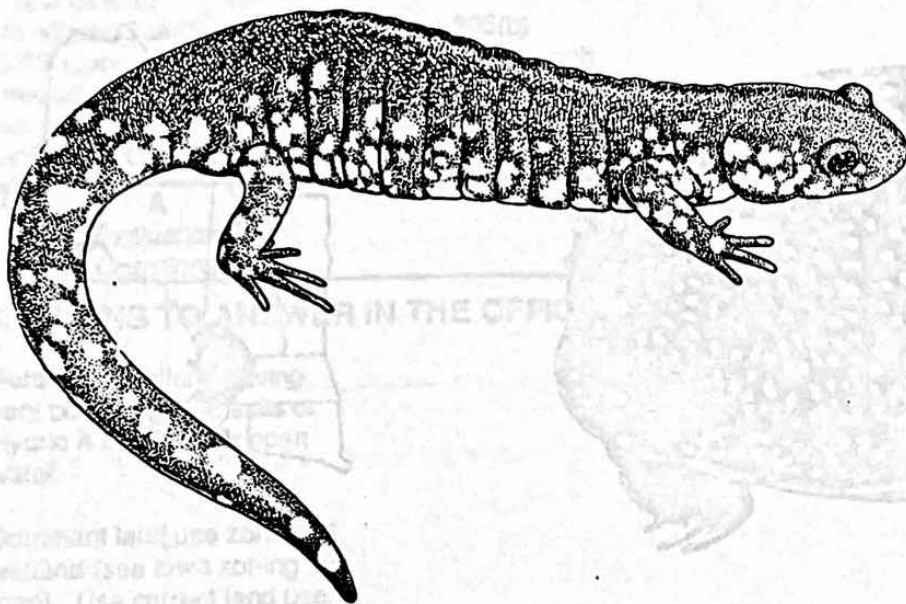
OTHER REFERENCES: Anderson and Gorman 1987; Smith 1961; Mason 1954.

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Blue-spotted Salamander

(*Ambystoma laterale*)



NOTE: See COMMENTS section for Jefferson salamander.

RANGE: Southeastern Quebec and the n. shore of the Gulf of St. Lawrence to James Bay and the s. end of Lake Winnipeg, s. to n. Illinois and Indiana, n. New York and New England. Disjunct colonies in New Jersey, Long Island, Iowa, and Labrador.

RELATIVE ABUNDANCE IN NEW ENGLAND: Rare; threatened in southern portion of range.

HABITAT: Wooded, swampy or moist areas (Minton 1954). Occasionally in overgrown pastures. Sometimes occurs where soil is sandy, and may be found under logs or other forest debris (in hardwood forests in the remnants of glacial Lake Passaic in New Jersey) (Anderson and Giacosis 1967). Occurs in a wide range of elevations (in western Connecticut) and along the Connecticut River floodplain (M. Klemens, personal communication).

SPECIAL HABITAT REQUIREMENTS: Ponds or semi-permanent water for breeding.

AGE/SIZE AT SEXUAL MATURITY: Snout to vent length of 47 to 55 mm in Indiana (Minton 1954).

BREEDING PERIOD: During early spring rains when night temperatures are above freezing (Lazell 1968).

EGG DEPOSITION: March to early April. Eggs laid on the bottoms of temporary shallow forest ponds, roadside drainage ditches, temporary pasture ponds, kettle holes (Landre 1980), attached to litter or in bottom detritus (Stille 1954), and twigs (Uzzell 1976b:48.1).

NO. EGGS/MASS: Varies greatly: 199 to 247 eggs (Uzzell 1964); 82 to 489 (Minton 1972:36); 6 to 10 eggs per mass (Landre 1980); often laid singly (Uzzell 1967b).

TIME TO HATCHING: About 1 month (Smith 1961:28).

LARVAL PERIOD: Extending to late June or mid-August (Smith 1961:28).

HOME RANGE/MOVEMENT: Unreported.

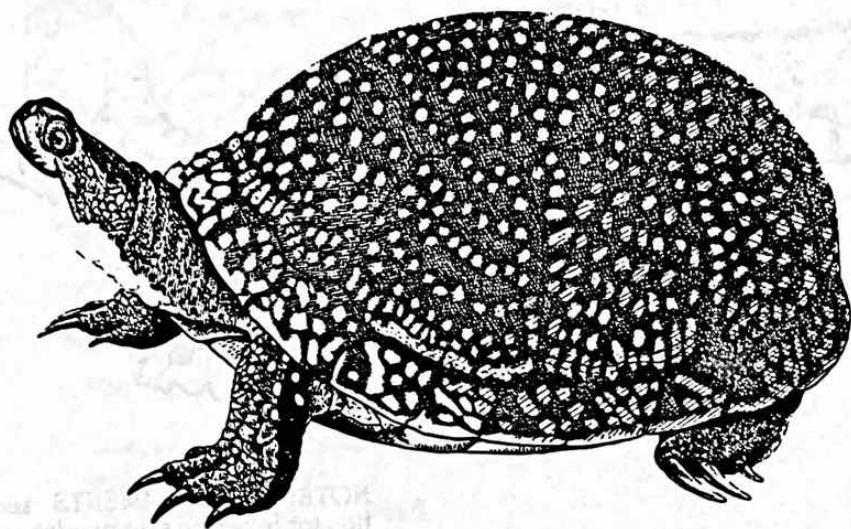
FOOD HABITS/PREFERENCES: Arthropods, annelids, and centipedes.

COMMENTS: Acid precipitation and habitat loss are major threats to this species in the Northeast.

KEY REFERENCES: Anderson and Giacosis 1967; Bleakney 1957; Landre 1980; Uzzell 1964, 1967b.

Blanding's Turtle

(*Emydoidea blandingii*)



RANGE: Scattered colonies in New York, New Hampshire, and e. Massachusetts. Southern Quebec across the Lake States to c. Minnesota, s. to Iowa and c. Illinois. Spotted occurrence from Nova Scotia to Ohio.

RELATIVE ABUNDANCE IN NEW ENGLAND: Populations localized and distribution spotty throughout its range (McCoy 1973:136.1). Generally scarce to rare, locally abundant in Massachusetts (Lazell 1972). An endangered species in Canada.

HABITAT: Shallow waters preferred; marshes, bogs,itches, ponds, swamps, also in protected coves and inlets of large lakes with abundant aquatic vegetation. May wander overland. Basks on logs, stumps, banks. Active in winter or hibernates in mud or debris.

SPECIAL HABITAT REQUIREMENTS: Shallow waters with soft muddy bottoms and aquatic vegetation.

AGE/SIZE AT SEXUAL MATURITY: During 12th year for males with a plastron length of 181 to 190 mm, Massachusetts (Graham and Doyle 1977); males 131 to 190 mm in Michigan (Gibbons 1968b). Size differences between these two populations probably due to differences in food quality and availability (Graham and Doyle 1977).

BREEDING PERIOD: Early spring through October, most from March to May (Ernst and Barbour 1972:181). Nesting in late April (T. Graham, personal communication).

EGG DEPOSITION: June to July. Nests made in sandy soils of upland areas.

CLUTCH SIZE: 6 to 11 eggs (Carr 1952:136), typically 8 to 9 eggs, clutches of 9, 13, and 16 eggs for Massachusetts females (T. Graham, personal communication). Clutch of 17 for a July nesting female (Graham and Doyle 1979). Two clutches may be laid each season.

INCUBATION PERIOD: Unreported.

EGGS HATCH: Autumn or next spring.

HOME RANGE/MOVEMENT: Less than 100 m for 4 individuals in a marsh in sw. Michigan (Gibbons 1968b).

FOOD HABITS/PREFERENCES: Crustaceans, insects, mollusks, fish, carrion, aquatic plants, succulent shoots, and berries. Crustaceans and crayfish account for about 50 percent of diet, insects more than 25 percent, and other invertebrates and vegetable matter 25 percent (Lagler 1943).

COMMENTS: Escaped individuals found in Connecticut (Lamson 1935). Primarily diurnal. In Michigan found in rivers (M. Klemens, personal communication).

KEY REFERENCES: Gibbons 1968b, Graham and Doyle 1977, 1979, McCoy 1973.

Wetland Name/Code: _____

NEEDED FOR THIS EVALUATION:

- Zoning map
- SCS soils map
- N.H. Water Quality Report to Congress 305(b)
- USGS topographic map or recent aerial photograph
- A method to calculate area (Dot grid, planimeter, etc.)
- Ruler or scale
- Map wheel (Optional)

Functional Value 1 ECOLOGICAL INTEGRITY

A Evaluation Questions	B Computations or Actual Value	C Evaluation Criteria	D Functional Value Index (FVI)
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QUESTIONS TO ANSWER IN THE OFFICE:

- | | | |
|---|---|-------------------|
| 1. Percent of wetland having very poorly drained soils or Hydric A soils and/or open water. | ☒ a. More than 50 percent
☐ b. From 25 to 50 percent
☐ c. Less than 25 percent | 1.0
0.5
0.1 |
| 2. Dominant land use zoning of wetland (see town zoning map). Use current land use if different from what is zoned. | ☐ a. Agriculture, forestry, or similar open space zoning
☒ b. Rural residential
☐ c. Commercial/industrial, high density residential | 1.0
0.5
0.1 |

QUESTIONS TO ANSWER IN THE FIELD:

- | | | |
|--|---|-------------------|
| 3. Water quality of the water-course, pond, or lake associated with the wetland. | ☒ a. High: Minimal pollution. Actual water quality meets or exceeds Class A or B standards
☐ b. Medium: Moderate pollution. Actual water quality is below Class B standards | 1.0
0.5 |
| 4. Ratio of the number of occupied buildings within 500 feet of the wetland edge to the total area of the wetland (acres). | ☐ a. Less than 1 bldg: 10 acres (<0.10)
☒ b. From 1 bldg: 10 acres to 1 bldg: 2 acres (0.10-0.50)
☐ c. More than 1 bldg: 2 acres (>0.5) | 1.0
0.5
0.1 |
| 5. Percent of original wetland filled. | ☐ a. Less than 10 percent
☒ b. From 10 to 50 percent
☐ c. More than 50 percent | 1.0
0.5
0.1 |
| 6. Percent of wetland edge bordered by a buffer of woodland or idle land at least 500 feet in width. | ☐ a. More than 80 percent
☒ b. From 20 to 80 percent
☐ c. Less than 20 percent | 1.0
0.5
0.1 |
| 7. Level of human activity WITHIN WETLAND as evidenced by litter, bike trails, roads, residences, etc. | ☒ a. Low level: Few trails in use and/or sparse litter
☐ b. Moderate level: Some used trails, roads, etc.
☐ c. High level: Many trails, roads, etc. within wetland | 1.0
0.5
0.1 |

Continued on next page...

Wetland Name/Code: _____

NEEDED FOR THIS EVALUATION:

- USGS topographic map
- Land use map and/or recent aerial photographs
- Ruler or scale
- A method to calculate area (Dot grid, planimeter, etc.)
- N.H. Water Quality Report to Congress 305(b)

Functional Value 2 WETLAND WILDLIFE HABITAT

A	B	C	D
Evaluation Questions	Computations or Actual Value	Evaluation Criteria	Functional Value Index (FVI)

QUESTIONS TO ANSWER IN THE OFFICE:

1. Ecological integrity.

Average FVI from Functional Value 1 2.5

2. Area of shallow permanent open water (less than 6 feet deep) including streams in or adjacent to wetland.

- | | |
|------------------------|-----|
| a. More than 3 acres | 1.0 |
| b. From 0.5 to 3 acres | 0.5 |
| c. Less than 0.5 acre | 0.1 |

QUESTIONS TO ANSWER IN THE FIELD:

3. Water quality of the watercourse, lake, or pond associated with the wetland.

FVI from Question V.1.3 1.0

4. Wetland diversity.

- | | |
|--|-----|
| a. Three or more wetland classes present | 1.0 |
| b. Two wetland classes present | 0.5 |
| c. One wetland class present | 0.1 |

5. Dominant wetland class.

- | | |
|--|-----|
| a. Emergent marsh and/or shallow open water | 1.0 |
| b. Forested and/or scrub-shrub wetland | 0.5 |
| c. Scrub-shrub saturated (bog) or wet meadow | 0.1 |

6. Interspersion of vegetation classes and/or open water.

- | | |
|--|-----|
| a. At least two wetland classes highly interspersed. Areas of each class scattered within wetland like a patchwork quilt | 1.0 |
| b. Moderate interspersion of wetland classes | 0.5 |
| c. Low degree of interspersion. Each wetland class is more or less contiguous and separate from the other classes | 0.1 |

Inventory of the Surrounding Lands

Hoveys Pond Conservation Area

for the

1995

Russell T. Hopping

July 25, 1995

Boxford Trails Association / Boxford Open Land Trust

70 Acres

West Boxford, MA

The Boxford Trails Association/Boxford Open Land Trust (BTA/BOLT) has been reasonably successful in its efforts to protect some of the remaining open space in Boxford and the public access to these lands. As BTA/BOLT continues its land protection mission, it must not lose sight of its land management obligation with its existing properties. In an effort to address land management issues BTA/BOLT adopted a set of Guidelines for Drafting Management Plans for BTA/BOLT Properties in April 1995. These guidelines were developed to provide BTA/BOLT with a focus in developing its property management and becoming good land stewards.

Under the inventory section, (item #3) of the management guidelines, lands surrounding BTA/BOLT properties are identified as one of the resources that should be inventoried. One of the objectives of inventorying surrounding lands is to identify the relationship BTA/BOLT properties have with their surrounding area. Property integrity may be influenced or depend on some or all of the surrounding lands. Abutters to BTA/BOLT properties are in the best position to support the well being, and BTA/BOLT efforts in general, of these properties. Inversely, abutters can also be responsible for unwanted intrusions onto BTA/BOLT properties. In some cases BTA/BOLT properties may be entirely surrounded by developed lands in which case the surrounding lands represent contacts and potential supporters which may influence property management through interest and involvement (i.e. property monitoring, presence as deterrent, donations, etc.) as well as a potential source for unwanted activities. In other cases opportunities for expansion or the establishment of buffers may exist with undeveloped lands. In either case knowledge of the status and ownership of the surrounding lands may facilitate the management of BTA/BOLT lands. Planning, an important part of land management, will be facilitated by knowing the status of abutting lands and how these lands influence BTA/BOLT properties. Ecological and recreational (i.e. public access) integrity may depend on abutting lands. In addition, BTA/BOLT lands, together with surrounding lands, may collectively represent valuable viewsheds or scenic areas important

to the sense of Boxford as a rural, country community. Preservation of these areas will rely on cooperative management to achieve BTA/BOLT goals. An inventory that identifies the ownership and land use of the surrounding lands of BTA/BOLT properties will serve as a resource for the Trust that will facilitate property management as described above.

serve as a stimulus and nucleus for a Greater Hoveys Pond Conservation Area which would protect Hoveys Pond and the surrounding fields and woodlands. This area has high wildlife, recreational, agricultural, and scenic value. This property currently represents the greatest management responsibility for BTA/BOLT. The Hoveys Pond property receives the greatest public use out of BTA/BOLT's current holdings and much of this use includes activities either not encouraged by BTA/BOLT or not permitted (i.e. hunting, ATVs and motorcycles). Information regarding surrounding lands has been gathered and compiled from the Boxford assessors office. This inventory will be included in the property's management plan and used to facilitate land management at this property.

63A - State tax break program for agricultural land

R - Residential lot

Open Land - Tax Exempt Land

Ind - Industrial Land

Open Land - Tax rate for "back" land (usually land locked)

ConsCom - Land owned by the Town's Conservation Commission

The following is an inventory of the lands surrounding the 70 acre Hoveys Pond Conservation Area in West Boxford, MA. The property was acquired by BTA/BOLT in the spring of 1994 for the purpose of protecting habitat for several rare species which had been identified as occurring on the property. In addition, it was hoped this property would serve as a stimulus and nucleus for a Greater Hoveys Pond Conservation Area which would protect Hoveys Pond and the surrounding fields and woodlands. This area has high wildlife, recreational, agricultural, and scenic value. This property currently represents the greatest management responsibility for BTA/BOLT. The Hoveys Pond property receives the greatest public use out of BTA/BOLT's current holdings and much of this use includes activities either not encouraged by BTA/BOLT or not permitted (i.e. hunting, ATVs and motorcycles). Information regarding surrounding lands has been gathered and compiled from the Boxford assessors office. This inventory will be included in the property's management plan and used to facilitate land management at this property.

ConsCom - Land owned by the Town's Conservation Commission

[illegible]

Hoveys Pond Conservation Area Abutters

Map#	Block #	Parcel #	Acres	Owner(s)	Address	Land Use
9	1	1A	27.25	Ruth C. Nason	Lake Shore Rd. P.O. Box 44	61A
9	1	3	13.8	Frederick H. Glore	285 Washington St. P.O. Box 341	R
9	1	4	1	Frederick H. Glore	285 Washington St. P.O. Box 341	R
9	1	5	11.2	Ruth C. Nason	Washington St. P.O. Box 44	61A
9	1	6	8.17	Brooks School	1160 Great Pond Rd. N. Andover	EXM Land
9	1	7	1.76	New England Power Co.	25 Research Dr. Westborough, MA	IND Land
9	1	7.1	2	Joseph & Lois Perkins	232 Washington St. P.O. Box 206	R
9	1	8	2	Patrick & Ann Saitta	220 Washington St. P.O. Box 143	R
9	1	10	18.68	John F. G. Eichom Jr.	1531 Georgina Ave. Santa Monica, CA	61A
9	1	19	3.33	Ruby B. Gould	181 Washington St. P.O. Box 505	R
9	1	20	1.127	Alan & Sharon Benson	181 Washington St. P.O. Box 13	R
9	1	22	6	William & Iva & William Jr.	Washington St. P.O. Box 124	R
9	1	23	18	Richard & Eva King	44 Belvedere Rd.	61A
9	1	25	2.81	Susan Richards	258 Washington St. P.O. Box 158	R
9	1	26	2.03	Gerald & Cynthia Masek	225 Washington St. P.O. Box 466	R
9	1	27.1	3.14	Micheal & Janice McGrath	269 Washington St.	R
9	1	28A	42.25	Nason	Washington St. P.O. Box 44	61A
9	1	28C	7.75	Nason	Washington St. P.O. Box 44	R
9	1	29	17	Albion A. Bergstrom	12700 Knightsbridge Dr. Woodbridge, VA	Open Land
9	1	30	4.9	Nancy A. Pearl	170 Lakeshore Rd. P.O. Box 172	R
9	1	30.1	2.008	Joseph Buturlia & Jean White	170 Lakeshore Rd. P.O. Box 474	R
9	1	31	2.001	Francisco & Delila Serret	172 Lakeshore Rd. P.O. Box 155	R
9	1	32	2	Raymond & Carolyn Mansell	251 Washington St.	R
9	1	35	6	John & Margaret Cleary	267 Washington St. P.O. Box 492	R
9	1	35.1	2	Michael Hendrickson	265 Washington St.	R
9	1	36	5.14	Carrie Barker	1267 Osgood St. N. Andover 01845	R
9	1	37	2.06	Placido & Donna Laudani	221 Washington St.	R
9	1	38	0.69	George Barker Jr.	1267 Osgood St. N. Andover 01845	R
9	1	39	2.77	Carrie M. Barker	1267 Osgood St. N. Andover 01845	R
9	1	40	3.37	Carrie M. Barker	1267 Osgood St. N. Andover 01845	R

9	2	9	49.5	New England Power Co.	25 Research Dr. Westborough, MA 01582	IND Land
9	2	10	1.75	Richard P. Chadwick	562 Boxford Rd. Bradford, MA 01830	R
9	2	11	2	Joseph & Mary Cassano	15 Belvedere Rd. P.O. Box 288	R
9	2	12	2	Joyce P. Young	19 Belvedere Rd.	R
9	2	13	2.01	Peter Piazza & Deborah Bowers	11 Plummer Rd. P.O. Box 453	R
9	2	14	2.94	Anne Phelan & Kathryn Donahue	15 Plummer Rd.	R
9	2	15	2.03	Lottie E. Narath	16 Plummer Rd. P.O. Box 247	R
9	2	16	2	Anthony & Gillian Mark	8 Plummer Rd. P.O. Box 85	R
9	2	17	2	Richard & Ritta Carroll	33 Belvedere Rd.	R
9	2	18	2	Rachelle Selkouits	41 Belvedere Rd. P.O. Box 908	R
9	2	19	2.01	William Clancey	43 Belvedere Rd. P.O. Box 262	R
9	2	20	2	William & Margaret Wheeler	42 Belvedere Rd. P.O. Box 493	R
9	2	21		David & Frances Davis		R
9	2	34	6.9	Richard & Eva King	44 Belvedere Rd.	R

5	1	1	2.1	Nancy Boisselle	156 Lakeshore Rd.	R
5	1	1.1	1.04	Nancy A. Pearl	170 Lakeshore Rd. P.O. Box 172	Open Land
5	1	2	2.5	William Markey	150 Lakeshore Rd. P.O. Box 365	R
5	1	3	2.7	Reginald & Marjorie Scribner	146 Lakeshore Rd. P.O. Box 211	R
5	1	4	3.5	Bradford & Diana Howes	138 Lakeshore Rd.	R
5	1	5	4.4	Nancy A. Pearl	170 Lakeshore Rd. P.O. Box 172	61A
5	1	6	3.4	Nancy A. Pearl	170 Lakeshore Rd. P.O. Box 172	61A
5	1	7	39.2	Gordon & Susan Price	22 Washington St.	61A
5	1	8	6.2	Damiano & Rita & Filippo Scarfo	93 Leyden St. East Boston, MA 02128	R
5	1	8.1	9.3	Town of Boxford		ConsCom
5	1	8.2	2.2	Town of Boxford		ConsCom

